

درس سیستم عامل

تمرین عملی اول سوال 1

متین باقری

فاطمیما تیمارچی

(i) (i)

در این تمرین من از Debian 13 استفاده کردم .

ابتدا به حالت `sudo` با استفاده از دستور `su` - رفتیم و سپس دستور گفته شده را در ترمینال زدیم .

[illegible][illegible]

ابتدا زیپ فایل مورد نظر رو از گیت هاب دانلود کردم و سپس آن را unzip کردم .

```
fatiMa-t@debian:~/Downloads$ ls
xv6-riscv-riscv.zip
fatiMa-t@debian:~/Downloads$ unzip xv6-riscv-riscv.zip
Archive:  xv6-riscv-riscv.zip
 90b2575ae6ef4d0927fedb2425af5c54ffa23df
   creating: xv6-riscv-riscv/
  inflating: xv6-riscv-riscv/.dir-locals.el
  inflating: xv6-riscv-riscv/.editorconfig
  inflating: xv6-riscv-riscv/.gdbinit.tpl-riscv
  inflating: xv6-riscv-riscv/.gitignore
  inflating: xv6-riscv-riscv/LICENSE
  inflating: xv6-riscv-riscv/Makefile
  inflating: xv6-riscv-riscv/README
   creating: xv6-riscv-riscv/kernel/
  inflating: xv6-riscv-riscv/kernel/bio.c
  inflating: xv6-riscv-riscv/kernel/buf.h
  inflating: xv6-riscv-riscv/kernel/console.c
  inflating: xv6-riscv-riscv/kernel/defs.h
  inflating: xv6-riscv-riscv/kernel/elf.h
  inflating: xv6-riscv-riscv/kernel/entry.S
  inflating: xv6-riscv-riscv/kernel/exec.c
  inflating: xv6-riscv-riscv/kernel/fcntl.h
  inflating: xv6-riscv-riscv/kernel/file.c
  inflating: xv6-riscv-riscv/kernel/file.h
  inflating: xv6-riscv-riscv/kernel/fs.c
  inflating: xv6-riscv-riscv/kernel/fs.h
  inflating: xv6-riscv-riscv/kernel/kalloc.c
  inflating: xv6-riscv-riscv/kernel/kernel.ld
  inflating: xv6-riscv-riscv/kernel/kernelvec.S
  inflating: xv6-riscv-riscv/kernel/log.c
  inflating: xv6-riscv-riscv/kernel/main.c
  inflating: xv6-riscv-riscv/kernel/memlayout.h
  inflating: xv6-riscv-riscv/kernel/param.h
  inflating: xv6-riscv-riscv/kernel/pipe.c
  inflating: xv6-riscv-riscv/kernel/plic.c
  inflating: xv6-riscv-riscv/kernel/print.c
  inflating: xv6-riscv-riscv/kernel/proc.c
  inflating: xv6-riscv-riscv/kernel/proc.h
  inflating: xv6-riscv-riscv/kernel/riscv.h
  inflating: xv6-riscv-riscv/kernel/sleeplock.c
  inflating: xv6-riscv-riscv/kernel/sleeplock.h
  inflating: xv6-riscv-riscv/kernel/spinlock.c
  inflating: xv6-riscv-riscv/kernel/spinlock.h
  inflating: xv6-riscv-riscv/kernel/start.c
  inflating: xv6-riscv-riscv/kernel/stat.h
  inflating: xv6-riscv-riscv/kernel/string.c
  inflating: xv6-riscv-riscv/kernel/switch.S
  inflating: xv6-riscv-riscv/kernel/syscall.c
  inflating: xv6-riscv-riscv/kernel/syscall.h
  inflating: xv6-riscv-riscv/kernel/sysfile.c
  inflating: xv6-riscv-riscv/kernel/sysproc.c
  inflating: xv6-riscv-riscv/kernel/trampoline.S
  inflating: xv6-riscv-riscv/kernel/trap.c
  inflating: xv6-riscv-riscv/kernel/types.h
  inflating: xv6-riscv-riscv/kernel/uart.c
  inflating: xv6-riscv-riscv/kernel/virtio.h
```

سپس با استفاده از دستور make qemu آن را کامپایل کردم .

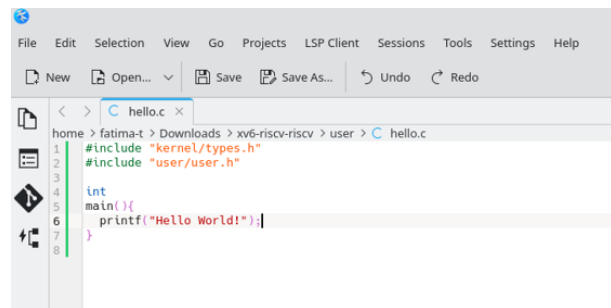
```
xv6-riscv-riscv: make - Konsole
- bash - Downloads/xv6-riscv-riscv: make
fatiMa-t@debian:~/Downloads/xv6-riscv-riscv$ make qemu
gcc -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/user.o user/userlib.c
perl user/sys.pl > user/sys.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/print.o user/print.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/umalloc.o user/umalloc.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/cat.o user/cat.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_cat user/cat.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_cat > user/cat.asm
riscv64-linux-gnu-objdump -S user_cat | sed '1,/SYMBOL TABLE/d; s; . * // ; / %d' > user/cat.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/echo.o user/echo.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_echo user/echo.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_echo > user/echo.asm
riscv64-linux-gnu-objdump -S user_echo | sed '1,/SYMBOL TABLE/d; s; . * // ; / %d' > user/echo.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/forktest.o user/forktest.c
# forktest has less library code linked in - needs to be small
# in order to be able to max out the proc table.
riscv64-linux-gnu-ld -z max-page-size=4096 -N -e main -o user_forktest user/forktest.o user/userlib.o user/sys.o
riscv64-linux-gnu-ld: warning: user_forktest has a LOAD segment with MAX permissions
riscv64-linux-gnu-objdump -S user_forktest > user/forktest.asm
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/grep.o user/grep.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_grep user/grep.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_grep > user/grep.asm
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/init.o user/init.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_init user/init.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_init > user/init.asm
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/kill.o user/kill.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_kill user/kill.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_kill > user/kill.asm
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/ln.o user/ln.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_ln user/ln.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_ln | sed '1,/SYMBOL TABLE/d; s; . * // ; / %d' > user/ln.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -D -fno-omit-frame-pointer -gdb -gdwarf-2 -march=rv64gc -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pr
inf -fno-builtin-printf -fno-builtin-printf -fno-builtin-printf -fno-stack-protector -fno-pie -no-pie -c -o user/mv.o user/mv.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_mv user/mv.o user/userlib.o user/sys.o user/print.o user/umalloc.o
riscv64-linux-gnu-objdump -S user_mv > user/mv.asm
```

داشته باشیم

[illegible][illegible]

فایل های تغییر یافته به صورت زیر هستند (فایل های تغییر یافته را در همین پوشه قرار داده ام):

در ابتدا باید یک فایل به نام `hello.c` بسازیم :



سپس باید در فایل `user.h` خط `void hello(void)` را اضافه کنیم :

```
46
47 // umalloc.c
48 void* malloc(uint);
49 void free(void*);
50
51 void hello(void);
52
```

در فایل `makefile` نیز باید خط `$U/_hello\` را در `UPROGS` اضافه می‌کنیم .

```
UPROGS=\
    $U/_cat\
    $U/_echo\
    $U/_forktest\
    $U/_grep\
    $U/_init\
    $U/_kill\
    $U/_ln\
    $U/_ls\
    $U/_mkdir\
    $U/_rm\
    $U/_sh\
    $U/_stressfs\
    $U/_usertests\
    $U/_grind\
    $U/_wc\
    $U/_zombie\
    $U/_logstress\
    $U/_forphan\
    $U/_dorphan\
    $U/_hello\
```

در نهایت هم با نوشتن `hello` در ترمینال عبارت `Hello World!` چاپ می‌شود :

```
fatima-tedeblan:~/Downloads/xv6-riscv-riscv$ make qemu
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=off
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ hello
Hello World!
$
```